ModelIt!

Teacher's Guide

Why Does a Black Car Get Hotter?

How Surface Color Affects Heat Absorption from Sunlight

Quick Reference

Grade Level	Grade 04L2
NGSS Standard	4-PS3-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 4-PS3-2

Make observations to provide evidence that energy can be transferred from place to place by sound, light, thermal energy, and electric currents.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a five-component computational model showing how sunlight intensity and surface color interact to determine heat absorption, surface temperature, and comfort level.
Disciplinary Core Idea	**PS3.B Conservation of Energy and Energy Transfer** — Energy can be transferred from place to place by light. When light energy is absorbed by a surface, it is converted to thermal energy. Dark surfaces absorb more light energy than light surfaces.
Crosscutting Concept	**Cause and Effect** — Students identify causal relationships between surface color and temperature, understanding that darker colors cause more heat absorption because they absorb rather than reflect light energy.

CAST Testing Alignment

MULTIPLE CHOICE:

Two identical metal plates are placed in the sun for 30 minutes. One is painted black and the other is painted white. The black plate reaches 140 degrees F while the white plate reaches only 95 degrees F. Which best explains this difference?

CONSTRUCTED RESPONSE:

Your school is building a new outdoor eating area in a very sunny location. Using what you know about light energy, surface color, and heat absorption, write a recommendation to the principal about what color to paint the tables and benches. Use the words 'light energy,' 'heat absorption,' and 'surface temperature' in your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

The relationship between color and heat is something we experience every day but rarely think about deeply. From the color of our clothes to the design of buildings and cities, understanding how surfaces interact with light energy helps engineers solve real problems like urban heat islands, energy-efficient buildings, and even spacecraft thermal protection.

How Sunlight Becomes Heat

Sunlight carries energy in the form of electromagnetic waves. When these waves hit a surface, one of two things happens: the energy is absorbed (taken in by the surface) or reflected (bounced back). Absorbed light energy transforms into thermal energy, raising the surface temperature. The more light energy a surface absorbs, the hotter it gets. This is the same process that warms the entire Earth — our planet absorbs sunlight and converts it to heat.

Why Dark Colors Absorb More

Color is actually about which wavelengths of light a surface reflects back to our eyes. A red ball looks red because it reflects red light and absorbs all other colors. A black surface absorbs nearly ALL wavelengths of light, converting all that energy to heat. A white surface reflects nearly ALL wavelengths, sending the energy away. This is why black absorbs the most heat and white absorbs the least — it is not magic, it is physics. The more light a surface reflects, the less energy it keeps as heat.

Urban Heat Islands

Cities are often 5 to 10 degrees hotter than surrounding rural areas because of the 'urban heat island' effect. Dark asphalt roads, dark rooftops, and concrete buildings absorb enormous amounts of sunlight and convert it to heat. This extra heat makes cities uncomfortable and increases energy costs for air conditioning. Environmental engineers combat this by using 'cool roofs' (white or reflective roofing), lighter-colored pavement, and increasing tree cover for shade. Some cities have reduced temperatures by several degrees just by changing surface colors.

Color and Comfort in Daily Life

Understanding heat absorption affects many everyday decisions. Light-colored clothing keeps you cooler in summer because it reflects sunlight. Car manufacturers know that dark-colored cars require more air conditioning energy to cool down. Desert animals often have lighter fur or skin to reflect sunlight. Even the color of your phone case matters — a black phone case in direct sun can get hot enough to trigger the phone's thermal shutdown. Engineers use this knowledge to design everything from solar panels (which WANT to absorb light) to spacesuits (which need to reflect it).

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify Sunlight Intensity and Surface Color as external components (environmental conditions and design choices) and Heat Absorption, Surface Temperature, and Comfort Level as internal components that respond to the externals.

E **Establish Relationships**

Students determine that Sunlight Intensity positively affects Heat Absorption, Surface Color (dark) positively affects Heat Absorption, Heat Absorption positively affects Surface Temperature, Surface Temperature negatively affects Comfort Level, and Sunlight Intensity positively affects Surface Temperature.

V **Visualize & Model**

Students build a five-component model showing how light energy flows from the sun through surface color to heat absorption, raising surface temperature and decreasing comfort level.

E **Evaluate Outcomes**

Students run four scenarios (bright/dark, bright/light, cloudy/dark, cloudy/light) to observe how sunlight intensity and surface color combine to determine surface temperature and comfort.

R **Revise & Extend**

Students add wind speed, surface material, or shade coverage to explore why some dark surfaces are still comfortable (shade) and why some light surfaces are still hot (metal material conducts heat quickly).

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Has anyone ever burned their hand on a car seat in the summer? Why was it so hot? And why are some cars worse than others?"*

Watch For

Ask students to share experiences touching hot surfaces in the sun. Prompt: Was the hot surface dark or light colored?

Slide 2: Learning Objectives

3 min

Say: *"Today we are going to figure out exactly WHY color matters when it comes to heat. It is all about what happens to sunlight when it hits a surface."*

Watch For

Have students read objectives aloud. Pre-teach absorption vs. reflection using a flashlight on black and white paper.

Slide 3: Big Question

3 min

Say: *"Same sun, same parking lot, same amount of time. But one car is burning hot and the other is comfortable. Where does the extra heat come from?"*

Watch For

Turn and talk: Why do you think the black car is hotter? Write predictions in notebooks.

Slide 4: LEVER Framework

2 min

Say: *"We are building a model with two things we can control — how much sun there is and how dark the surface is — and three things that respond."*

Watch For

Preview the five components. Hold up black and white paper and ask: Which do you think will get hotter in the sun?

Slide 5: Activity 1: Components

8 min

Say: *"Sunlight Intensity and Surface Color are our external components — they are the conditions we set. Heat Absorption, Surface Temperature, and Comfort Level are the results."*

Watch For

Guide component sorting. Discuss why comfort is internal — we do not choose how comfortable a surface feels, the temperature determines it.

Slide 6: Activity 2: Connections

10 min

Say: *"Here is the key: dark surfaces INCREASE heat absorption, and more heat absorption INCREASES temperature. But higher temperature DECREASES comfort. See how the system connects?"*

Watch For

Map relationships together. Emphasize the negative relationship between temperature and comfort.

Slide 7: Activity 3: Simulation

12 min

Say: *"Watch what happens when we change the surface from dark to light. Look at how much the temperature drops!"*

Watch For

Run all four scenarios. Have students record temperatures and comfort levels for each combination.

Slide 8: Discoveries

5 min

Say: *"Dark surfaces are heat sponges — they soak up sunlight and turn it into heat. Light surfaces are like mirrors — they bounce the sunlight away."*

Watch For

Demonstrate with a flashlight: shine it on black paper (light disappears) vs. white paper (light bounces visibly). The disappeared light became heat!

Slide 9: STEM Challenge

12 min

Say: *"Your mission: design the coolest possible parking lot for a school in a super hot city. Use your model data to make smart choices about color!"*

Watch For

Distribute materials (colored paper, thermometers, lamp). Groups test surfaces and design their cool parking lot with a poster explaining their choices.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Sunlight Intensity, Surface Color
- **INTERNAL:** Heat Absorption, Surface Temperature, Comfort Level

Relationships

- **Sunlight Intensity to Heat Absorption:** POSITIVE (+) — Stronger sunlight delivers more light energy to the surface. More energy hitting the surface means more energy available to be absorbed and converted to heat. A surface in direct midday sun absorbs far more energy than the same surface on a cloudy day.
- **Surface Color (dark) to Heat Absorption:** POSITIVE (+) — Darker surfaces absorb a higher percentage of the light energy that hits them. A black surface absorbs about 95% of incoming light, while a white surface absorbs only about 20%. More absorption means more light energy converting to thermal energy.
- **Heat Absorption to Surface Temperature:** POSITIVE (+) — When more light energy is absorbed and converted to thermal energy, the surface temperature rises. The relationship is direct — double the absorption and the temperature increase roughly doubles. Surfaces that absorb the most energy get the hottest.
- **Surface Temperature to Comfort Level:** NEGATIVE (-) — As surface temperature increases, comfort level decreases. A cool surface is comfortable to touch and sit on, while a hot surface can cause burns and discomfort. Beyond about 120 degrees F, surfaces become painful to touch.
- **Sunlight Intensity to Surface Temperature:** POSITIVE (+) — Stronger sunlight increases surface temperature both directly and indirectly through increased heat absorption. On the hottest, sunniest days, all surfaces reach their highest temperatures regardless of color.

Key Discoveries

- Dark surfaces absorb MORE light energy than light surfaces, converting that energy into heat that raises the surface temperature
- Light-colored surfaces REFLECT most of the sunlight away, so less energy becomes heat and the surface stays cooler
- Even a dark surface does not get extremely hot on a cloudy day because there is less light energy available to absorb
- Both sunlight intensity AND surface color matter together. The hottest surfaces happen when STRONG sunlight hits DARK colors, and the coolest surfaces happen when WEAK sunlight hits LIGHT colors

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Black objects are hotter because black is a 'hot color'"	Black is not inherently hot — it ABSORBS more light energy. A black surface in a completely dark room is the same temperature as a white surface. The difference only appears when light energy hits the surfaces. Black absorbs about 95% of incoming light, while white reflects about 80%. The absorbed light energy converts to thermal energy (heat). Remove the light source and the temperature difference disappears.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"White surfaces do not absorb any heat at all"	White surfaces DO absorb some light energy — typically about 20% of incoming sunlight. They are cooler than black surfaces, but they are not perfectly reflective. Only a perfect mirror would reflect 100% of light. White surfaces still warm up in the sun, just much less than dark ones. In very strong sunlight, even white surfaces can get uncomfortably warm.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"The color of a surface does not matter if it is in the shade"	This is actually MOSTLY true, which is why it is a useful misconception to address. In shade, there is very little direct sunlight to absorb, so color differences in temperature nearly disappear. This shows that the heat difference between dark and light surfaces is caused by SUNLIGHT, not by the colors themselves. However, some indirect light still reaches shaded surfaces, so a small temperature difference may remain.